Q. What is E-Commerce Security? Explain the Major Security Requirements of E-Commerce.

E-Commerce involves the exchange of **money, personal information, and business data** through electronic networks. Therefore, protecting online transactions and customer information from unauthorized access, fraud, and cyber threats is essential.

Major Security Requirements of E-Commerce

1. Confidentiality

Confidentiality ensures that sensitive information such as **passwords, payment details, and personal data** is accessible only to authorized users. Encryption is commonly used to prevent unauthorized persons from viewing confidential information.

2. Integrity

Integrity ensures that information remains **accurate, complete, and unchanged** during transmission or storage. It prevents attackers from modifying transaction details or other important business information.

3. Authentication

Authentication verifies the **identity of a user or party** involved in an online transaction. Methods such as passwords, OTPs, biometrics, and digital certificates can be used to confirm identity.

4. Authorization

Authorization determines **what an authenticated user is allowed to access or perform** within an E-Commerce system. For example, a normal customer should not have access to an administrator's account.

5. Availability

Availability ensures that E-Commerce websites, applications, and services remain **accessible to legitimate users whenever required**. Protection against attacks such as DoS and DDoS helps maintain continuous service availability.

6. Non-Repudiation

Non-repudiation ensures that a user cannot **deny a transaction or action that they have actually performed**. Digital signatures and electronic transaction records help provide this security.

Q. What is Wireless Communication? Explain the Major Security Issues Associated with Wireless Networks. (5 Marks)

Introduction

Wireless communication allows devices to exchange information without using physical cables. Technologies such as **Wi-Fi, Bluetooth, and mobile networks** provide mobility and convenience, but the transmission of data through the air also creates several security risks.

Security Issues Associated with Wireless Networks

1. Unauthorized Access

Poorly secured wireless networks can allow unauthorized users or attackers to gain access to the network. This may result in **data theft, misuse of network resources, and unauthorized activities**.

2. Data Interception (Eavesdropping)

Wireless signals can be intercepted by attackers within the communication range. If the data is not properly encrypted, sensitive information such as **usernames, passwords, and financial details** may be exposed.

3. Man-in-the-Middle Attack

In this attack, an attacker secretly positions themselves between two communicating parties and **intercepts or modifies the information** being exchanged. Users may remain unaware that their communication has been compromised.

4. Fake or Rogue Access Point

Attackers can create a **fake Wi-Fi hotspot** that appears to be a legitimate network. When users connect to it, attackers may monitor their activities or steal sensitive login and financial information.

5. Malware Attacks

An insecure wireless network can provide an opportunity for attackers to spread **viruses, worms, spyware, or ransomware** to connected devices. Such attacks may result in data loss and disruption of business operations.

6. Denial of Service (DoS) Attack

In a DoS attack, an attacker sends excessive traffic or interference to a wireless network, making it **slow or unavailable to legitimate users**. This can interrupt online services and E-Commerce operations.

Ways to Improve Wireless Security

- Use **strong and unique passwords** for wireless networks.
- Enable **WPA2 or WPA3 encryption**.
- Avoid performing sensitive transactions through unsecured public Wi-Fi.
- Keep operating systems, routers, and security software **regularly updated**.
- Use **firewalls and antivirus software**.
- Enable **Two-Factor Authentication (2FA)** for important accounts.

Q. Differentiate between B2B, B2C and C2C E-Commerce Models with Suitable Examples. (5 Marks)

Introduction

E-Commerce provides different ways of conducting business transactions through electronic networks. **B2B, B2C, and C2C** are three major E-Commerce models classified according to the relationship between the buyer and seller.

Meaning of B2B, B2C and C2C

B2B (Business-to-Business):

It refers to online transactions conducted **between two or more business organizations**, such as a manufacturer purchasing raw materials from a supplier.

B2C (Business-to-Consumer):

It refers to the online sale of **products or services directly from a business to the final consumer**, such as a customer buying a mobile phone from an online store.

C2C (Consumer-to-Consumer):

It refers to transactions in which **one consumer sells products or services directly to another consumer**, usually through an online marketplace.

Basis	B2B	B2C	C2C
Full Form	Business to Business	Business to Consumer	Consumer to Consumer
Parties Involved	Business ↔ Business	Business → Consumer	Consumer ↔ Consumer
Main Purpose	Conducts business-to-business transactions	Sells products/services to final consumers	Facilitates exchange between individual consumers
Nature of Transaction	Wholesale or business-oriented	Retail or consumer-oriented	Resale or exchange of personal goods
Transaction Volume	Usually high	Usually small or medium	Usually individual
Relationship	Long-term business relationship is common	Mainly seller-customer relationship	Mostly transaction-based relationship
Example	A manufacturer purchasing raw materials from a supplier	A customer buying a laptop from an online store	A person selling a used mobile phone to another person

Benefits of SEO

1. Increases Website Traffic

SEO helps a website appear in search engine results when users search for related information. As more people find the website, the number of visitors increases naturally without paid advertisements.

2. Improves Search Engine Ranking

SEO improves the quality and structure of a website, which helps it rank higher on search engines like Google. A higher ranking makes the website more visible to users.

3. Builds Trust and Credibility

Websites that appear on the first page of search results are usually trusted more by users. Good content, fast loading speed, and secure browsing (HTTPS) also improve credibility.

4. Provides Long-Term Marketing Benefits

Unlike paid advertisements, SEO continues to bring visitors even after the optimization work is done. This makes it a cost-effective and long-term marketing strategy.

5. Increases Sales and Business Growth

When more targeted visitors come to a website, the chances of selling products or services increase. This helps improve business revenue and supports long-term business growth.

Basis	TPS	MIS	DSS	ESS
Full Form	Transaction Processing System	Management Information System	Decision Support System	Executive Support System
Users	Operational staff	Middle-level managers	Managers and analysts	Top-level executives
Purpose	Records daily business transactions.	Provides reports for management.	Helps solve complex problems and make decisions.	Supports strategic planning and long-term decisions.
Decision Level	Operational level	Tactical (Managerial) level	Semi-structured decisions	Strategic level
Example	Billing system, ATM, payroll system	Sales report, inventory report	Sales forecasting, budgeting	Business growth planning, market analysis

Q. What is Encryption? Explain Symmetric and Asymmetric Key Encryption. (5 Marks)

Introduction

E-Commerce transactions involve sensitive information such as **passwords, banking details, and personal data**. Encryption protects this information by converting it into an unreadable form so that unauthorized users cannot understand it.

Definition of Encryption

Encryption is the process of converting readable data, known as **plaintext**, into an unreadable form, known as **ciphertext**, using an encryption algorithm and a key.

Decryption is the reverse process of converting ciphertext back into its original readable form.

Types of Encryption

1. Symmetric Key Encryption

Symmetric encryption uses **the same secret key for both encryption and decryption**. The sender and receiver must securely share the same key before communication.

Example: AES and DES.

Advantages:

- It is **fast and efficient**.
- It is suitable for encrypting **large amounts of data**.

Limitation: Securely sharing the secret key between sender and receiver can be difficult.

2. Asymmetric Key Encryption

Asymmetric encryption uses a **pair of keys: a public key and a private key**. The public key can be shared openly, while the private key is kept secret by its owner.

Example: RSA.

Advantages:

- It provides a safer method of **key exchange**.
- It supports **authentication and digital signatures**.

Limitation: It is generally **slower** than symmetric encryption.

Q. What is Digital Signature? Explain its Requirements and Working Mechanism. (5 Marks)

A **Digital Signature** is a cryptographic technique used to verify the **authenticity, integrity, and identity of the sender** of a digital message or document. It also provides **non-repudiation** to electronic transactions.

Requirements of Digital Signature

1. Authentication

Digital signature verifies the **identity of the sender or signer**. It helps the receiver confirm that the message actually came from the claimed sender.

2. Integrity

It ensures that the message or document has **not been altered or modified** after it was signed. Any change in the original data can make the signature invalid.

3. Non-Repudiation

It prevents the sender from **denying a transaction or message** that they have digitally signed. This provides accountability in electronic transactions.

4. Security

A digital signature uses **cryptographic algorithms and private keys** to protect the signature from being forged or misused.

Working Mechanism of Digital Signature

Step 1: Creation of Message

The sender prepares the **digital message or document** that needs to be transmitted.

Step 2: Hash Generation

A **hash function** is applied to the message to generate a unique fixed-length value called a **message digest**.

Step 3: Signature Creation

The sender uses their **private key** to encrypt/sign the message digest. This creates the **digital signature**.

Step 4: Transmission

The sender sends the **original message along with the digital signature** to the receiver.

Step 5: Verification

The receiver uses the sender's **public key** to verify the digital signature and generates a new hash of the received message. If both values match, the message is considered **authentic and unchanged**.

Conclusion

A digital signature acts as a **digital proof of identity and authenticity** in electronic communication. It provides **authentication, integrity, and non-repudiation**, making it an important security mechanism for E-Commerce transactions.

Q. What is E-Commerce Security? Discuss the Major Security Threats and Defense Strategies Used to Secure E-Commerce Systems. (10 Marks)

E-Commerce Security refers to the technologies, policies, procedures, and practices used to **protect E-Commerce systems, users, transactions, and data from unauthorized access, fraud, modification, theft, and cyber attacks**.

Major Security Threats in E-Commerce

1. Phishing

Phishing is a fraudulent technique in which attackers create **fake emails, messages, or websites** that appear to be from legitimate organizations. Users may be tricked into revealing passwords, OTPs, or payment information.

2. Malware Attacks

Malware includes **viruses, worms, Trojan horses, spyware, and ransomware** that can infect devices and systems. It may steal information, damage files, or disrupt E-Commerce operations.

3. Data Theft

Attackers may attempt to steal sensitive information such as **customer details, passwords, credit card information, and business data**. Data theft can cause financial loss and serious damage to customer trust.

4. Man-in-the-Middle Attack
In a Man-in-the-Middle attack, an attacker secretly intercepts communication between two parties. The attacker may **monitor or modify sensitive information** without the knowledge of the sender or receiver.
5. Denial-of-Service Attack
A DoS or DDoS attack attempts to make an E-Commerce website unavailable by sending **a large amount of traffic or requests**. This can interrupt online services and cause business losses.
6. Password Attacks
Weak or reused passwords can be attacked through methods such as **brute-force and dictionary attacks**. If an attacker obtains a customer's password, they may gain unauthorized access to the account.
7. Web Application Attacks
Vulnerabilities in websites and applications can be exploited to gain unauthorized access or manipulate data. Examples include **SQL Injection and Cross-Site Scripting (XSS)**.
Defense Strategies for E-Commerce Security
1. Encryption
Encryption converts readable information into an **unreadable form** so that unauthorized persons cannot understand it. It is widely used to protect sensitive information during online transactions.
2. Authentication and Authorization
Strong authentication methods such as **passwords, OTPs, biometrics, and multi-factor authentication** help verify users. Authorization ensures that users can access only the resources they are permitted to use.
3. Digital Signature and PKI
Digital signatures help provide **authentication, integrity, and non-repudiation**. Public Key Infrastructure (PKI) manages digital certificates and cryptographic keys to establish trusted electronic communication.
4. Firewall and Intrusion Detection System
A **firewall** monitors and controls network traffic according to security rules. An **Intrusion Detection System (IDS)** monitors activities and alerts administrators about suspicious or unauthorized behavior.
5. Regular Software Updates
Web servers, operating systems, applications, and security tools should be **regularly updated and patched**. Updates help remove known security vulnerabilities that attackers could exploit.
6. Backup and Recovery
Important business and customer data should be **regularly backed up**. Proper backup and recovery procedures help an organization restore its services and information after ransomware, system failure, or other attacks.
7. User Awareness and Security Training
Employees and customers should be educated about **phishing, suspicious links, weak passwords, and online fraud**. Human awareness is an important defense because many attacks exploit user mistakes.

Q. What is Wireless Communication? Explain Different Wireless Technologies Used in E-Commerce and Discuss the Security Issues Associated with Wireless Communication. (10 Marks)
Introduction
Wireless communication enables devices to exchange **data and information without physical cables**. It has become an important part of E-Commerce because customers can access websites, make payments, use mobile applications, and communicate with businesses from almost anywhere.
Definition of Wireless Communication
Wireless communication is the process of transmitting **voice, data, or information between two or more devices through electromagnetic signals without using a physical wired connection**.
Wireless Technologies Used in E-Commerce
1. Wi-Fi
Wi-Fi provides wireless Internet connectivity within a limited area such as homes, offices, restaurants, and shopping centers. It allows customers to **browse websites, place orders, and access online services** conveniently.
2. Bluetooth
Bluetooth is a short-range wireless technology used to exchange data between nearby devices. It can support **mobile devices, payment-related devices, and other smart business equipment**.
3. WLAN
Wireless Local Area Network (WLAN) connects devices within a limited geographical area using wireless communication. Businesses can use WLAN to connect employees, computers, and other devices without extensive cabling.
4. WiMAX
WiMAX provides wireless broadband connectivity over a **larger geographical area** than traditional Wi-Fi. It can be used to provide Internet access where wired broadband infrastructure is limited.
5. Mobile Communication
Mobile technologies such as **3G, 4G, and 5G** provide Internet connectivity through cellular networks. They enable customers to use mobile shopping applications, online banking, digital payments, and other E-Commerce services while travelling.
6. WAP
Wireless Application Protocol (WAP) is a communication protocol designed to provide Internet-based services through mobile devices. It enables users to access certain web-based services through wireless networks.
Security Issues Associated with Wireless Communication
1. Unauthorized Access
Poorly protected wireless networks can allow unauthorized users or attackers to gain access. This may result in **data theft, misuse of network resources, and privacy violations**.
2. Data Interception
Wireless signals travel through the air and can be intercepted by attackers. Without proper encryption, sensitive information such as **passwords, personal details, and financial data** may be exposed.
3. Fake or Rogue Access Points
Attackers can create **fake Wi-Fi hotspots** that look like legitimate networks. When users connect to these networks, attackers may capture their credentials and other confidential information.
4. Man-in-the-Middle Attack
An attacker may secretly position themselves between two communicating parties and **intercept or modify the exchanged information**. This can be particularly dangerous during online transactions.
5. Malware Attacks
An insecure wireless network can become a channel for spreading **viruses, worms, spyware, and ransomware**. Such attacks may damage systems and interrupt E-Commerce operations.
6. Denial-of-Service Attack
Attackers may overload a wireless network with excessive traffic or interference, making the network **slow or unavailable to legitimate users**. This can interrupt online services and cause business losses.
Measures to Improve Wireless Security
- Use **strong and unique passwords** for wireless networks.
- Enable **WPA2 or WPA3 encryption**.
- Use secure authentication and **Two-Factor Authentication (2FA)**.
- Keep routers, operating systems, and applications **regularly updated**.
- Avoid sensitive transactions through **unsecured public Wi-Fi**.
- Use firewalls and antivirus software to protect connected devices.

Conclusion
Wireless communication has made E-Commerce **more mobile, flexible, and accessible**, but it also introduces several security challenges. Therefore, the use of strong **encryption, authentication, secure network configurations, regular updates, and user awareness** is essential for maintaining safe and reliable wireless E-Commerce operations.

Q. What are TPS, MIS, DSS and ESS? Explain their Roles in an Organization. (10 Marks)
1. Transaction Processing System (TPS)
Transaction Processing System (TPS) is a computer-based system that records and processes the **routine day-to-day transactions** of an organization.
Role of TPS
- Records transactions such as **sales, purchases, payroll, and banking**.
- Maintains accurate and up-to-date transaction records.
- Provides basic data required by other information systems.
- Improves the **speed and efficiency** of routine business operations.

Example: A supermarket's billing system that records every customer purchase.
2. Management Information System (MIS)
Management Information System (MIS) collects and processes organizational data to provide **useful and structured information to managers** for planning, monitoring, and controlling business activities.
Role of MIS
- Converts transaction data into **meaningful management reports**.
- Helps managers monitor organizational performance.
- Supports **planning and controlling** activities.
- Provides information for routine and structured decision-making.

Example: A monthly sales report showing the sales performance of different branches.
3. Decision Support System (DSS)
Decision Support System (DSS) is a computer-based system that helps managers **analyze information, evaluate alternatives, and make decisions**, especially in semi-structured or complex situations.
Role of DSS
- Supports **problem-solving and decision-making**.
- Performs data analysis and comparisons.
- Helps managers evaluate different alternatives.
- Supports techniques such as **what-if analysis and forecasting**.

Example: A company using sales data to predict future demand and decide how much inventory to maintain.
4. Executive Support System (ESS)
Executive Support System (ESS) provides top-level executives with **summarized and strategic information** required for long-term planning and decision-making.
Role of ESS
- Provides an overview of the organization's **overall performance**.
- Helps executives identify important **trends and opportunities**.
- Supports strategic and long-term decision-making.
- Presents information through **dashboards, graphs, summaries, and Key Performance Indicators (KPIs)**.

Example: A CEO viewing a dashboard showing revenue, profit, market growth, and overall business performance.

Q. What is E-Commerce? Explain the Major E-Commerce Models with Suitable Examples. (10 Marks)
Introduction
E-Commerce has changed the traditional way of buying and selling by allowing business activities to be performed through **electronic networks and the Internet**. It enables businesses, consumers, and other organizations to exchange products, services, money, and information electronically.
Definition of E-Commerce
Electronic Commerce (E-Commerce) refers to the buying, selling, marketing, payment, and exchange of **goods, services, and information through electronic networks, particularly the Internet**.
Major E-Commerce Models
1. B2B (Business-to-Business)
B2B refers to online transactions conducted **between two or more business organizations**. Businesses use this model to purchase raw materials, products, software, and other services from other businesses.
Example: A restaurant purchasing food supplies from an online wholesaler.
2. B2C (Business-to-Consumer)
B2C refers to the direct sale of **products or services from a business to the final consumer** through an online platform. It is one of the most common E-Commerce models.
Example: A customer purchasing a laptop from an online retail store.
3. C2C (Consumer-to-Consumer)
C2C allows **one consumer to sell products or services directly to another consumer**, generally through an online marketplace.
Example: A person selling a used mobile phone to another person through an online marketplace.
4. C2B (Consumer-to-Business)
C2B is a model in which an **individual consumer offers products, services, or value to a business**. The business becomes the buyer or customer.
Example: A freelance graphic designer providing design services to a company.
5. B2G (Business-to-Government)
B2G refers to electronic transactions between **business organizations and government agencies**. Businesses may provide products, services, or solutions to government organizations through online systems.
Example: A software company providing an online management system to a government office.
6. G2C (Government-to-Citizen)
G2C refers to the electronic delivery of **government services and information directly to citizens** through online platforms.
Example: Citizens applying for government services or paying taxes through an online government portal.

Q. What is SEO? Explain the SEO Cycle and Discuss the Role of Bots in SEO. (10 Marks)
Introduction
In E-Commerce, having a good website is not enough if potential customers cannot find it through search engines. **Search Engine Optimization (SEO)** helps improve the visibility and ranking of a website so that it can attract more relevant visitors and potential customers.
Definition of SEO
Search Engine Optimization (SEO) is the process of improving a website's **content, structure, technical performance, and online presence** to achieve better visibility and ranking in search engine results and attract organic traffic.
SEO Cycle
SEO is a continuous process rather than a one-time activity. The major steps of the SEO cycle are:
1. Keyword Research
The first step is to identify the **words and phrases that potential users search for**. Selecting relevant keywords helps a website target the right audience.
2. Website Analysis
The website is examined to identify problems related to **content, structure, speed, mobile-friendliness, links, and technical performance**. This helps determine areas that require improvement.
3. On-Page Optimization
The website's content and HTML elements are optimized using relevant **keywords, headings, titles, meta descriptions, images, and internal links**. This makes the content easier for both users and search engines to understand.
4. Off-Page Optimization
Activities are performed outside the website to improve its **authority and reputation**. Building quality backlinks, promoting content, and maintaining an online presence are common off-page SEO activities.
5. Search Engine Crawling and Indexing
Search engine bots visit the website, discover its pages, and analyze their content. Relevant pages are then stored in the search engine's **index** so they can appear in search results.
6. Monitoring and Analysis
Website performance is continuously monitored using factors such as **traffic, ranking, clicks, and conversions**. The results help identify what is working and what needs improvement.
7. Improvement and Re-optimization
Based on performance and changes in search engine algorithms, the website is **updated and optimized again**. Thus, SEO becomes a continuous cycle of improvement.

SEO Cycle in Simple Form

Keyword Research → Website Analysis → On-Page SEO → Off-Page SEO → Crawling & Indexing → Monitoring → Improvement → Again SEO

Role of Bots in SEO

1. Crawling Websites

Search engine bots automatically visit websites and **discover new or updated web pages**.

2. Collecting Information

Bots analyze information such as **text, headings, links, images, and other page elements** to understand the content of a webpage.

3. Indexing Pages

After analyzing a webpage, bots help search engines store relevant information in their **search index**. Indexed pages can then appear in search results.

4. Discovering New Content

When a website publishes new pages or updates existing content, bots can discover those changes through links, sitemaps, and repeated crawling.

5. Evaluating Website Structure

Bots examine links and website structure to understand the **relationship between different pages** and determine which content can be accessed and indexed.

Conclusion

SEO is a **continuous optimization process** that helps an E-Commerce website gain visibility, organic traffic, and potential customers. The SEO cycle involves **research, optimization, crawling, monitoring, and improvement**, while search engine bots play a key role in **discovering, analyzing, and indexing web content**.

Q. What is Electronic Payment System? Explain its Types, Benefits and Limitations. (10 Marks)

Introduction

The growth of E-Commerce has made electronic payment an essential part of online business. Customers can purchase goods and services and make payments **electronically without using physical cash**.

Definition of Electronic Payment System

An **Electronic Payment System (EPS)** is a system that enables individuals and organizations to **transfer money and make payments electronically** through digital networks such as the Internet and banking systems.

Types of Electronic Payment Systems

1. Credit and Debit Cards

Credit and debit cards allow customers to make online payments by entering their **card details through a secure payment gateway**. They are widely used for online shopping and service payments.

2. Electronic Fund Transfer (EFT)

EFT enables money to be transferred electronically from **one bank account to another** without using physical cash or cheques. It is commonly used for business and personal transactions.

3. Electronic Wallet (E-Wallet)

An e-wallet is a digital application that allows users to **store payment information and make electronic transactions** using a mobile phone or other device.

4. Internet Banking

Internet banking allows customers to make payments and transfer money through their **bank's online platform**. Users can perform transactions without physically visiting a bank.

5. Mobile Payment

Mobile payment allows users to make payments using **mobile phones and mobile applications**. It is commonly used for shopping, bill payment, ticket booking, and other services.

6. Electronic Cheque

An electronic cheque is a **digital version of a traditional cheque** that can be used to authorize electronic payment through a computerized system.

Benefits of Electronic Payment System

1. Convenience

Customers can make payments **anytime and from almost anywhere** using an Internet-connected device. This makes online shopping and other E-Commerce activities more convenient.

2. Faster Transactions

Electronic payments are processed much faster than many traditional payment methods. This allows businesses to **receive payments and complete transactions quickly**.

3. Reduced Use of Cash

Electronic payment reduces the need to carry and handle **physical cash**, making transactions easier and more efficient.

4. Supports E-Commerce Growth

Online payment systems make it possible for customers to **purchase products and services online**, supporting the growth of E-Commerce businesses.

5. Better Transaction Records

Electronic payments automatically create **digital records of transactions**. These records are useful for accounting, tracking expenses, and financial management.

Limitations of Electronic Payment System

1. Security Risks

Electronic payment systems may face threats such as **phishing, hacking, identity theft, and payment fraud**. Weak security can result in financial and data losses.

2. Internet Dependency

Most electronic payment systems require a **stable Internet or network connection**. Network failure can interrupt or delay transactions.

3. Technical Problems

Server failures, software errors, system downtime, or device problems can prevent users from completing transactions successfully.

4. Privacy Concerns

Electronic transactions involve the collection and processing of **personal and financial information**, which may create privacy concerns if data is not properly protected.

5. Lack of Digital Awareness

People who have limited knowledge of digital technologies may find electronic payment systems **difficult to use or understand**, especially in areas with low digital literacy.

Conclusion

Electronic Payment Systems have become a **backbone of modern E-Commerce** by making transactions faster, convenient, and cashless. However, security, privacy, technical problems, and Internet dependency remain important challenges. Therefore, **secure payment gateways, strong authentication, encryption, and user awareness** are necessary for reliable electronic payments.

Q. Explain the Major Steps Involved in Developing an E-Commerce Website. (10 Marks)

Introduction

An E-Commerce website provides an online platform where customers can **search, compare, purchase, and pay for products or services**. A successful E-Commerce website should not only look attractive but also be **secure, user-friendly, fast, and reliable**.

Major Steps in Developing an E-Commerce Website

1. Requirement Analysis

The first step is to identify the **business goals, target customers, products, required features, and system requirements**. Proper requirement analysis provides a clear direction for website development.

2. Website Planning

In this stage, the overall structure and functionality of the website are planned. Important features such as **product catalog, search, shopping cart, user accounts, payment, order management, and customer support** are identified.

3. Domain and Hosting Selection

A suitable **domain name** is selected to provide the website with a unique online identity. Reliable web hosting is then selected to store website files and make the website accessible through the Internet.

4. Database Design

A database is designed to store important information such as **customer details, product information, prices, inventory, orders, and payment records**. A well-designed database ensures efficient storage and retrieval of information.

5. User Interface Design

The website interface is designed to provide a **simple, attractive, and user-friendly experience**. Important elements include navigation menus, product images, search options, shopping cart, and clear checkout pages.

6. Website Development

The planned design and functionality are converted into a working website using appropriate **frontend, backend, and database technologies**. Features such as registration, login, product management, cart, and order processing are implemented.

7. Payment Gateway Integration

A secure payment system is integrated so customers can pay electronically using available payment methods. The payment process should protect **customer and financial information**.

8. Security Implementation

Security measures such as **SSL/TLS encryption, secure authentication, strong passwords, firewalls, input validation, and regular updates** are implemented to protect the website from cyber threats.

9. Testing

The website is thoroughly tested to identify errors and ensure that all features work correctly. **Functionality, usability, security, compatibility, performance, and payment processes** should be tested before launch.

10. Deployment and Maintenance

After successful testing, the website is deployed to a live server and made available to customers. Regular **maintenance, security updates, backups, performance monitoring, and feature improvements** are required to keep the website reliable.

Q. What is Mobile Commerce (M-Commerce)? Explain its Characteristics, Advantages and Limitations. (10 Marks)

Introduction

The rapid growth of smartphones and mobile Internet has changed the way people purchase goods and services. **Mobile Commerce** allows customers to perform E-Commerce activities directly through mobile devices, making online business more convenient and accessible.

Definition of M-Commerce

Mobile Commerce (M-Commerce) refers to the buying, selling, payment, and exchange of **goods, services, and information through mobile devices such as smartphones and tablets using wireless networks**.

Characteristics of M-Commerce

1. Mobile Accessibility

M-Commerce allows users to access products and services through **mobile devices from almost anywhere**. Physical presence at a shop is not required.

2. Convenience

Customers can **search, compare, order, and pay for products** using their smartphones. This makes shopping quick and convenient.

3. Location Independence

Users can perform transactions while **travelling, working, or staying at different locations**, as long as mobile network or Internet access is available.

4. Personalization

Mobile applications can provide **personalized recommendations, offers, notifications, and services** based on user preferences and activities.

5. Real-Time Communication

Businesses can communicate with customers instantly through **push notifications, SMS, mobile applications, and instant messaging**.

6. Mobile Payment

M-Commerce supports various digital payment methods such as **mobile wallets, QR payments, mobile banking, and card-based payments**.

Advantages of M-Commerce

1. Greater Convenience

Customers can shop and make payments **anytime and almost anywhere** using their mobile devices. This saves time and effort.

2. Wider Customer Reach

Businesses can reach customers beyond physical locations through **mobile applications and mobile-friendly websites**. This helps expand the potential market.

3. Faster Transactions

Mobile applications provide quick access to **product information, ordering, and payment facilities**, making transactions faster.

4. Personalized Services

Businesses can use customer data to provide **personalized recommendations, discounts, and promotional offers**, improving customer experience.

5. Increased Business Opportunities

M-Commerce creates new opportunities for businesses to sell products and services through **mobile applications, social media, and mobile advertising**.

Limitations of M-Commerce

1. Security and Privacy Risks

Mobile transactions may face threats such as **phishing, malware, account theft, and unauthorized access**. Poor security can expose sensitive customer information.

2. Dependence on Internet and Mobile Networks

M-Commerce requires a reliable **Internet or mobile network connection**. Poor connectivity can interrupt browsing, ordering, or payment.

3. Small Screen Size

Compared with computers, mobile devices generally have **smaller screens**, which can make viewing detailed product information or completing complex tasks less convenient.

4. Device Compatibility Issues

Different mobile devices, operating systems, and screen sizes may cause **compatibility and performance problems** for mobile applications and websites.

Applications of M-Commerce

- **Mobile shopping** – purchasing products through mobile apps.
- **Mobile banking** – checking accounts and transferring money.
- **Mobile payments** – paying through wallets and QR codes.
- **Mobile ticketing** – purchasing bus, airline, movie, or event tickets.
- **Location-based services** – receiving offers and services based on the user's location.

Conclusion

M-Commerce has made E-Commerce **more mobile, personal, and convenient** by bringing online business directly to customers' smartphones. Although security, connectivity, and compatibility remain challenges, the growing use of mobile devices makes M-Commerce an important part of the **modern digital economy**.